

UNDERSTANDING THE ROLE OF REASONING DATA IN PRETRAINING

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ABSTRACT

Reasoning data expose models to multi-step reasoning, but a reasoning trace may reveal only a subset of the underlying facts it composes. How should pretraining balance atomic facts with reasoning traces? We study this question by constructing a setting in which every fact provides an atomic training signal, while reasoning traces contain only a subset of the facts they compose, enabling us to measure generalization to unseen transitions. For a linear transformer, we derive the weight assigned to atomic training signals that minimizes held-transition squared error and show that its leading-order optimum increases with reasoning-chain length and decreases with the fraction of facts held out from reasoning traces. These trends persist in a full transformer layer with an MLP, softmax attention, and cross-entropy loss, as well as in a realistic mathematical reasoning setting. Together, our results provide a principled view of how reasoning-chain structure should shape the allocation of training data during pretraining.

1 INTRODUCTION

Reasoning traces teach models how to compose facts across multiple steps, while atomic examples provide broader coverage of those facts. Under a fixed training budget, these objectives compete: spending more data on reasoning traces leaves less budget for learning individual facts, while spending more on atomic examples provides broader factual coverage but less practice with composition. This raises a fundamental question for pretraining: **how should a model’s training budget be divided between atomic facts and reasoning traces?**

This question becomes especially important as reasoning data move into pretraining, rather than being reserved for the later post-training stage. Recent work has begun to examine how reasoning data should be incorporated into pretraining. Akter et al. (2026) study the effects of when reasoning data are introduced during training, as well as their diversity and quality, while Baek et al. (2026) investigate repeatedly sampled domain data during pretraining. Recent training recipes likewise incorporate instruction data and thinking traces into Olmo 3 midtraining (Olmo et al., 2025) and SFT-style examples into Nemotron 3 Nano pretraining (Blakeman et al., 2025). Together, these developments establish reasoning-oriented data as an increasingly important component of the pretraining mixture. Yet they leave open a basic question: **how should the optimal mixture depend on the structure and coverage of reasoning data?**

The tension is subtle because reasoning traces provide something atomic examples do not: they teach the model how to use knowledge across multiple steps. But reasoning traces can also be an expensive way to teach that knowledge. Longer traces consume more of a fixed training budget, while a limited collection of traces may cover only a subset of the facts needed at evaluation. Thus, longer reasoning can simultaneously make compositional training more valuable and make broad factual coverage more important. This suggests that the optimal allocation should depend on both **reasoning-chain length** and **reasoning-trace coverage**, namely, the fraction of facts that appear inside reasoning traces. Our goal is to characterize this allocation under a fixed training budget, where differences in

learning dynamics matter even when different data mixtures might eventually reach similar performance.

We study this question first in **state tracking**, a controlled setting for multi-step reasoning over a finite environment (Merrill et al., 2024; Merrill & Sabharwal, 2025; Forner et al., 2026; Huang et al., 2026c; Li et al., 2025). Starting from an initial state, the model receives a sequence of relations and must infer the state reached after executing the entire sequence. This formulation captures capabilities that arise in practical language-model tasks, including tracking entities through a narrative, reasoning over sequences of chess moves, and executing programs (Kim & Schuster, 2023; Merrill et al., 2024). It has consequently become a standard testbed for studying language-model reasoning, both theoretically (Liu et al., 2023; Merrill et al., 2024; Merrill & Sabharwal, 2025) and empirically (Zhang et al., 2022; Liu et al., 2023; Li et al., 2025).

Our construction makes the distinction between **learning facts** and **learning to compose them** explicit. Each state transition corresponds to an atomic fact. We include every transition as an atomic training example, but construct reasoning traces using only a subset of the transitions. The remaining transitions are *held transitions*: the model sees them individually during training, but never encounters them inside a reasoning trace. Thus, a held transition is not an unseen fact—the model has learned it atomically—but an unseen *use* of that fact within a reasoning chain. This construction cleanly separates factual coverage from reasoning composition and lets us ask whether learning to compose facts transfers to facts that have never been used in a reasoning trace.

We first explore this tradeoff empirically using a one-layer transformer with softmax attention and an MLP. We vary reasoning-chain length and reasoning-trace coverage. **Perhaps surprisingly, we find that longer reasoning chains favor allocating more training data to atomic facts, not less.** Intuitively, longer traces provide richer compositional training, but each trace also consumes more of the fixed budget, making broad factual coverage increasingly important. Conversely, when reasoning traces cover more of the available facts, reasoning data become relatively more valuable. We then test whether these effects extend beyond the controlled setting using FineMath (allal et al., 2025) and MathLake (Cai et al., 2025). These experiments examine analogous effects of reasoning length and data coverage, as well as the stage of training at which reasoning data are introduced.

To explain these trends, we turn to a linear transformer, a standard simplification for analyzing transformer learning dynamics (Zhang et al., 2024; Ahn et al., 2023; Huang et al., 2025). Atomic training examples directly reinforce transitions that reasoning traces may never contain, while reasoning traces train the model to compose transitions across multiple steps. We derive the atomic-data weight that minimizes squared prediction error on held transitions. To leading order, the optimal weight **increases with reasoning-chain length and decreases with the fraction of held transitions**. In other words, longer reasoning chains favor greater allocation to atomic facts, whereas broader reasoning-trace coverage makes reasoning data relatively more valuable. These theoretical predictions agree with the trends observed in both our controlled and realistic experiments.

Contributions. Our contributions are three-fold. First, we introduce a state-tracking setting that separates **factual coverage** from **reasoning composition**: all transitions are observed atomically, but only a subset appears inside reasoning traces. Using a one-layer softmax-MLP transformer, we show that an interior mixture of atomic and reasoning examples can outperform either extreme, and characterize how the preferred mixture changes with reasoning-chain length and reasoning-trace coverage. Second, for a linear-attention model, we derive an explicit mixture law whose optimum depends on reasoning-chain length and coverage, showing why longer reasoning chains can favor greater allocation to atomic facts. Third, we develop a fixed-token FineMath-MathLake protocol to study the same allocation question in a realistic setting, varying reasoning length, data coverage, and when reasoning data are introduced during training.

2 RELATED WORK

Reasoning data in pretraining. Recent work has increasingly explored the role of reasoning data during pretraining, rather than reserving them for post-training. Akter et al. (2026) systematically study how reasoning data should be allocated between pretraining and supervised fine-tuning, showing that reasoning examples introduced during pretraining provide benefits that cannot be fully recovered by shifting the same supervision to post-training. Wang et al. (2025) find that mathematical and long-chain-of-thought data during mid-training improve both base-model reasoning and subsequent reinforcement-learning performance, while Hatamizadeh et al. (2026) incorporate explicit reasoning directly into the pretraining objective. An earlier related observation comes from Allen-Zhu & Li (2024), who, although not studying reasoning data specifically, argue from controlled knowledge-learning experiments that instruction-finetuning data should be mixed into pretraining rather than reserved entirely for post-training. These works establish the value of introducing richer forms of supervision during pretraining, but leave open a more fundamental question: *how should reasoning data be balanced against atomic data when the training budget is fixed?* We study this tradeoff by separating factual coverage from reasoning composition: every fact is observed atomically, but only a subset is exposed within reasoning traces. This allows us to characterize when reasoning data are most valuable and how their benefit depends on reasoning-chain length and coverage.

Transformer theory. A growing body of work uses controlled synthetic tasks to understand how Transformers acquire, retrieve, and compose structured knowledge. Nichani et al. (2025) analyze factual recall through associative memories, showing how shallow Transformers can store and retrieve large collections of subject-relation-object associations. Huang et al. (2026b) study how attention learns to select relevant relations and retrieve corresponding facts from a pretrained associative memory. Most closely related to our setting, Huang et al. (2026c) prove that Transformers trained by gradient descent can learn chain-of-thought state tracking and characterize how the algebraic structure of the transition system governs length generalization through attention concentration. These works provide insight into how Transformers store and compose knowledge, but do not study how different forms of training data interact under a fixed budget. We instead consider a setting in which every transition is observed through atomic examples, while only a subset appears in reasoning traces. This cleanly separates **factual availability** from **exposure to multi-step composition**, allowing us to analyze how the allocation between atomic and reasoning data affects the ability to compose transitions that are known individually but absent from the reasoning data.

3 TOY SETTING: STATE TRACKING

3.1 PRELIMINARIES

Definition 1 (State-tracking world and held transitions). *Let $[K] = \{1, \dots, K\}$ and $N, M \geq 2$. Entities are indexed by $e \in [N]$, and each relation $r \in [M]$ acts through a fixed permutation $\pi_r : [N] \rightarrow [N]$. The full fact table $\mathcal{A} = [N] \times [M]$ contains $D = NM$ entity-relation pairs. A length- L chain $\chi = (r_1, \dots, r_L, e_0) \in [M]^L \times [N]$, with $L \geq 2$, induces the state sequence*

$$e_\ell = \pi_{r_\ell}(e_{\ell-1}), \quad \ell \in [L].$$

For an integer $1 \leq m < D$, draw the held set $\mathcal{H}$ and define its complement $\mathcal{S}$ by

$$\mathcal{H} \sim \text{Unif}\{H \subseteq \mathcal{A} : |H| = m\}, \quad q = \frac{m}{D}, \quad p = 1 - q, \quad \mathcal{S} = \mathcal{A} \setminus \mathcal{H}.$$

The held fraction q is the fraction of facts excluded from reasoning traces, and the reasoning coverage p is the fraction available to those traces. The mask is sampled once and fixed throughout each run.

Definition 2 (Atomic, reasoning, and evaluation populations). *An atomic example samples e_0, r_1 independently and uniformly and $r_2, \dots, r_L$ all belong to $\mathcal{S}$, but supervises only $e_1 = \pi_{r_1}(e_0)$. Its first entity-relation pair is therefore uniform over $\mathcal{A}$, including pairs in $\mathcal{H}$. A*

reasoning example is sampled uniformly from chains whose actual transitions all belong to $\mathcal{S}$ and supervises the complete state trace $e_1, \dots, e_L$. Formally,

$$\begin{aligned}\mathcal{D}_{\text{full}} &= \text{Unif}([M]^L \times [N]), \\ \mathcal{C}_L(\mathcal{S}) &= \{\chi \in [M]^L \times [N] : (e_{\ell-1}, r_\ell) \in \mathcal{S} \text{ for every } \ell \in [L]\}, \\ \mathcal{C}'_L(\mathcal{S}) &= \{\chi \in [M]^L \times [N] : (e_{\ell-1}, r_\ell) \in \mathcal{S} \text{ for every } \ell \in [2, L]\}, \\ \mathcal{D}_{\text{at}}(\mathcal{S}, L) &= \text{Unif}(\mathcal{C}'_L(\mathcal{S})) \\ \mathcal{D}_{\text{rea}}(\mathcal{S}, L) &= \text{Unif}(\mathcal{C}_L(\mathcal{S})).\end{aligned}$$

We write $\mathbb{E}_{\chi \sim \mathcal{D}_{\text{rea}}}$ for expectation over the reasoning population associated with the fixed set $\mathcal{S}$.

Learning objective. Entities and relations are represented by distinct typed identity tokens. Both data sources use the same relation-first input format:

$$\begin{aligned}\text{atomic:} \quad & [r_1, \dots, r_L, e_0] \longrightarrow [e_1], \\ \text{reasoning:} \quad & [r_1, \dots, r_L, e_0] \longrightarrow [e_1, \dots, e_L].\end{aligned}$$

The shared prefix places the first prediction at the same absolute position and supplies the same number of candidate relations in both sources. Atomic examples supervise only the first transition, whereas reasoning examples additionally supervise subsequent prediction roles. Every fact is available through atomic supervision, while reasoning traces cover only a subset, allowing us to study whether facts learned individually can be used within multi-step computations.

Let $\beta \in [0, 1]$ be the atomic *loss weight*. We average the reasoning loss over its L state predictions before combining the two sources:

$$\begin{aligned}\mathcal{L}_\beta^{\text{CE}}(\theta) &= \beta \mathbb{E}_{\chi \sim \mathcal{D}_{\text{at}}} \ell_{\text{CE},1}(\theta; \chi) \\ &\quad + (1 - \beta) \mathbb{E}_{\chi \sim \mathcal{D}_{\text{rea}}} \left[\frac{1}{L} \sum_{\ell=1}^L \ell_{\text{CE},\ell}(\theta; \chi) \right],\end{aligned}\tag{1}$$

where

$$\ell_{\text{CE},\ell}(\theta; \chi) = -\log p_\theta(e_\ell \mid [r_1, \dots, r_L, e_0, \dots, e_{\ell-1}]), \quad \ell \in [L].$$

Each cross-entropy term supervises the next entity using the true preceding states. A test chain is called *held* if at least one of its ground-truth transitions belongs to $\mathcal{H}$.

3.2 ONE FULL LAYER TRANSFORMER

Our learner consists of a single causal self-attention head followed by a position-wise multi-layer perceptron (MLP). We encode token identities and positions in separate one-hot blocks and directly parameterize the merged query-key matrix, following Zhang et al. (2024); Ahn et al. (2023); Huang et al. (2025; 2026c;b). We first define the input representation, followed by the attention and output maps.

Definition 3 (Token and positional encoding). *Let $u_r \in \mathbb{R}^M$ and $v_e \in \mathbb{R}^N$ be the standard basis vectors corresponding to relation r and entity e , respectively. Fix a maximum input-prefix length $T \geq 2L$, shared by atomic and reasoning examples. The token-content vectors are*

$$c(r) = \begin{pmatrix} u_r \\ 0_N \end{pmatrix}, \quad c(e) = \begin{pmatrix} 0_M \\ v_e \end{pmatrix}.$$

For a token x_t at position t , let $\mathbf{c}_t := c(x_t)$ and let $\mathbf{p}_t \in \mathbb{R}^T$ be the t -th standard basis vector. Concatenating content and position gives

$$\tilde{\mathbf{x}}_t = \begin{pmatrix} \mathbf{c}_t \\ \mathbf{p}_t \end{pmatrix} \in \mathbb{R}^d, \quad d = M + N + T, \quad \widetilde{\mathbf{X}}_t = [\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_t] \in \mathbb{R}^{d \times t}.$$

All content and positional vectors are fixed. Relation, entity, and position coordinates occupy disjoint blocks of sizes M , N , and T , respectively. We write $\mathbf{X}_t = [\mathbf{c}_1, \dots, \mathbf{c}_t] \in \mathbb{R}^{(M+N) \times t}$ for the corresponding content-only matrix.

Both data sources predict e_1 at position $L + 1$, and reasoning role ℓ predicts e_ℓ from the prefix ending at position $L + \ell$. The last supervised query is therefore at position $2L$, so $T \geq 2L$ covers all input prefixes. We reserve T for the maximum prefix length and use τ for the training horizon below.

We parameterize the merged query-key scores directly by a trainable matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$. In particular, the score assigned to key position i by query position j is $\tilde{\mathbf{x}}_i^\top \mathbf{A} \tilde{\mathbf{x}}_j$.

Definition 4 (Causal self-attention head). For $\mathbf{A} \in \mathbb{R}^{d \times d}$ and a prefix $\tilde{\mathbf{X}}_t \in \mathbb{R}^{d \times t}$, define

$$\text{attn}(\tilde{\mathbf{X}}_t; \mathbf{A}) := \tilde{\mathbf{X}}_t \text{softmax} \left(\text{MASK}(\tilde{\mathbf{X}}_t^\top \mathbf{A} \tilde{\mathbf{X}}_t) \right) \in \mathbb{R}^{d \times t},$$

where, for $S \in \mathbb{R}^{t \times t}$,

$$[\text{MASK}(S)]_{ij} = \begin{cases} S_{ij}, & i \leq j, \\ -\infty, & i > j. \end{cases}$$

The softmax is applied column-wise.

The attention output is passed through a shared MLP to produce next-token logits over the entity and relation vocabulary.

Definition 5 (One-layer attention-MLP Transformer). Fix an MLP width $m \geq 1$ and a coordinatewise nonlinearity ϕ . The trainable matrices are

$$\mathbf{A} \in \mathbb{R}^{d \times d}, \quad \mathbf{W} \in \mathbb{R}^{m \times d}, \quad \mathbf{V} \in \mathbb{R}^{m \times (M+N)},$$

with parameters $\boldsymbol{\theta}_{\text{TF}} = (\mathbf{A}, \mathbf{V}, \mathbf{W})$. At query position t , the vocabulary-logit vector is

$$f_{\text{TF},t}(\tilde{\mathbf{X}}_t; \boldsymbol{\theta}_{\text{TF}}) = \mathbf{V}^\top \phi \left(\mathbf{W} [\text{attn}(\tilde{\mathbf{X}}_t; \mathbf{A})]_{:,t} \right) \in \mathbb{R}^{M+N}.$$

The next-token distribution is

$$p_{\boldsymbol{\theta}_{\text{TF}}}(\cdot \mid \tilde{\mathbf{X}}_t) = \text{softmax} \left(f_{\text{TF},t}(\tilde{\mathbf{X}}_t; \boldsymbol{\theta}_{\text{TF}}) \right).$$

The first M output coordinates index relations, and the last N index entities. Accordingly, the probability assigned to entity $e \in [N]$ is component $M + e$ of this distribution.

We train this model using the mixed cross-entropy objective in equation 1.

Effect of data mixture Figure 1 shows that mixing atomic and reasoning examples improves OOD accuracy over either pure-source baseline, suggesting that the two forms of supervision play complementary roles. The empirically preferred atomic example ratio generally increases with reasoning length L and decreases with the held-out fact fraction q . This pattern is not strictly monotone across all selected grid maximizers: the $L \in \{3, 4\}$ cases at $q = 30\%$ deviate from the held-fraction trend. However, these panels exhibit near-optimal plateaus, with several mixture ratios attaining very similar mean OOD accuracy.

3.3 LINEAR-ATTENTION AND THE OPTIMAL MIXTURE

We next examine how the allocation between atomic examples and reasoning traces affects optimization. Our goal is to characterize the mixture that best supports the use of held transitions after a common training horizon, and to understand how this allocation depends on reasoning length and coverage. To make this interaction tractable, we study a scalar linear-attention surrogate

Following prior block-structured analyses of Transformer learning (Nichani et al., 2024; Huang et al., 2025; 2026a;b; Javanmard et al., 2026b;a), we use a block-diagonal attention score matrix with token-token and position-position blocks, fixing the mixed token-position blocks to zero. These two components contribute additively to the softmax logits, but their contributions multiply after exponentiation. This motivates combining the bilinear content scores with a learned positional gate through the Hadamard product in the linear surrogate:

$$(\mathbf{X}_t^\top \mathbf{B} \mathbf{X}_t) \odot \tilde{\mathbf{C}}_{1:t, 1:t}.$$

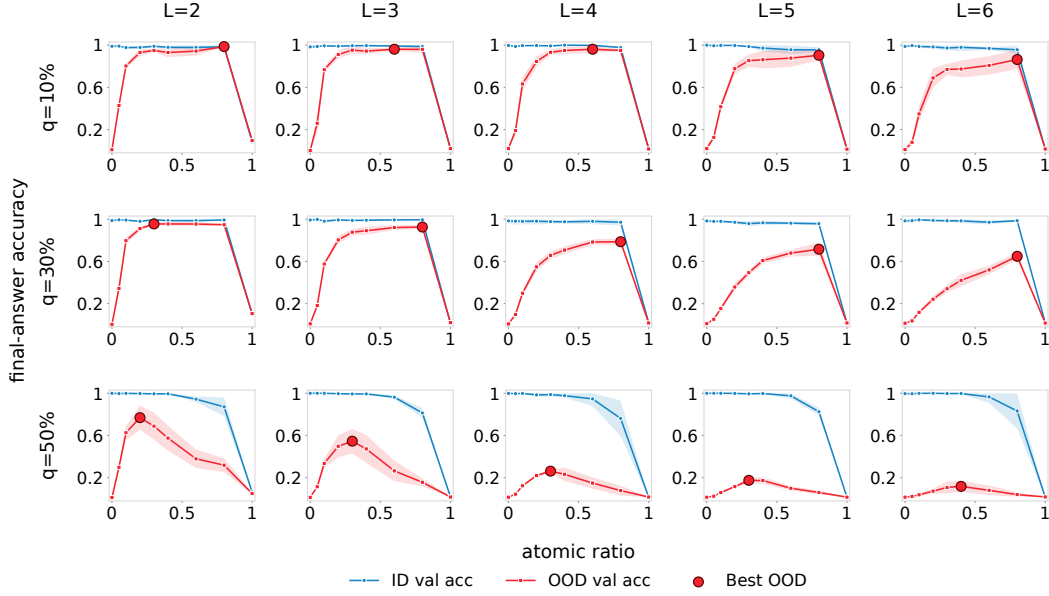


Figure 1: **Effect of atomic-reasoning mixing on state-tracking accuracy.** Final-answer accuracy as a function of the atomic ratio in a synthetic world with 64 entities and 8 relations, each sampled as an independent random permutation. All runs use 4000 AdamW updates with learning rate 0.002, weight decay 0.01, and batch size 256. Curves and shaded regions indicate the mean and ± 1 standard deviation across three training seeds. The empirically optimal atomic example ratio generally increases with reasoning length L and decreases with the held-out fact fraction q . The selected grid maximizers for $L \in \{3, 4\}$ at $q = 30\%$ deviate from the latter trend; however, these panels exhibit near-optimal plateaus, with several tested mixture ratios achieving very similar mean OOD accuracy.

Here $\mathbf{B}$ controls token interactions, while $\tilde{\mathbf{C}}$ controls positional routing. In particular, a zero gate entry excludes a connection, whereas an additive zero would leave its content score unchanged. Removing attention normalization and replacing the nonlinear output map with a scalar linear readout gives the following model.

Definition 6 (Scalar linear-attention model). Let $\mathbf{X}_t \in \mathbb{R}^{(M+N) \times t}$ contain the fixed typed one-hot content vectors, with $t \leq T$ and $T \geq 2L$. For $\mathbf{W} \in \mathbb{R}^{1 \times (M+N)}$, $\mathbf{B} \in \mathbb{R}^{(M+N) \times (M+N)}$, and $\tilde{\mathbf{C}} \in \mathbb{R}^{T \times T}$, define

$$f_{\text{lin}}(\mathbf{X}_t; \mathbf{W}, \mathbf{B}, \tilde{\mathbf{C}}) = \mathbf{W} \mathbf{X}_t \left[(\mathbf{X}_t^\top \mathbf{B} \mathbf{X}_t) \odot \tilde{\mathbf{C}}_{1:t, 1:t} \right] \in \mathbb{R}^{1 \times t}. \quad (2)$$

The trainable coordinates are the relation values $w_r = \mathbf{W}_{1,r}$, the relation-entity scores $b_{r,e} = \mathbf{B}_{r,M+e}$, and the routing coefficients $C_{j\ell} = \tilde{\mathbf{C}}_{j,L+\ell}$ for $j, \ell \in [L]$; all remaining entries are fixed at zero. Write $\theta = (w, b, C)$.

At role ℓ , the teacher-forced prefix ends at $e_{\ell-1}$ and has length $L + \ell$. Let $f_\ell(\chi; \theta)$ be entry $(1, L + \ell)$ of $f_{\text{lin}}(\mathbf{X}_{L+\ell}; \theta)$. For entity $e \in [N]$, define

$$\zeta_e = \frac{2e - N - 1}{N}, \quad \hat{e}_\ell \in \arg \min_{e \in [N]} |f_\ell(\chi; \theta) - \zeta_e|. \quad (3)$$

Squared loss and joint training. Atomic prefixes sample the initial entity and all relation labels independently and uniformly; reasoning prefixes are uniform over chains whose actual transitions belong to $\mathcal{S}$. For an atomic loss weight $\beta \in [0, 1]$, the mixed population

loss is

$$L_\beta(\theta; \mathcal{S}) = \frac{\beta}{2} \mathbb{E}_{\chi \sim \mathcal{D}_{\text{at}}} (f_1(\chi; \theta) - \zeta_{e_1})^2 + \frac{1-\beta}{2L} \sum_{\ell=1}^L \mathbb{E}_{\chi \sim \mathcal{D}_{\text{rea}}} (f_\ell(\chi; \theta) - \zeta_{e_\ell})^2. \quad (4)$$

Atomic examples supervise only the first state, whereas reasoning examples supervise the complete trace. Averaging the reasoning loss over its L predictions makes β the relative weight of the two source losses.

All active entries of $\mathbf{W}, \mathbf{B}, \tilde{\mathbf{C}}$ are jointly trained by gradient flow:

$$\dot{\theta}_\beta(t) = -\nabla_\theta L_\beta(\theta_\beta(t); \mathcal{S}), \quad \theta_\beta(0) = \theta_0. \quad (5)$$

We initialize $\mathbf{B}$ to zero and use the same initialization for every mixture:

$$\mathbf{B}(0) = 0, \quad w_r(0) = 1 \quad (r \in [M]), \quad C(0) = E_{11}. \quad (6)$$

Here E_{11} has a single unit entry at $(1, 1)$ and zeros elsewhere.

Held-transition error and optimal allocation. The training objective alone does not measure whether facts learned from atomic supervision can be used at later prediction roles. We therefore evaluate later transitions that were excluded from reasoning traces. For each $\ell \geq 2$, let

$$\mathcal{D}_{H,\ell} := \mathcal{D}_{\text{full}}(\cdot \mid (e_{\ell-1}, r_\ell) \in \mathcal{H}).$$

The conditioning event has probability $q > 0$, so this distribution is well-defined. Define the held-transition error by

$$\mathcal{E}_H(\theta) := \frac{1}{2(L-1)} \sum_{\ell=2}^L \mathbb{E}_{\chi \sim \mathcal{D}_{H,\ell}} (f_\ell(\chi; \theta) - \zeta_{e_\ell})^2.$$

At a training horizon $\tau > 0$, define the optimal atomic weight as

$$\beta_{\tau,S}^* \in \arg \min_{\beta \in [0,1]} \mathcal{E}_H(\theta_\beta(\tau)). \quad (7)$$

We now characterize how reasoning length and coverage affect the optimal atomic loss weight. The following theorem shows that, at a fixed training horizon, $\beta_{\tau,S}^*$ concentrates around a deterministic center $\Theta_{N,M}(L, 1-q)$ as the numbers of entities and relations grow. This center increases with reasoning length and decreases with the held fraction, describing how the preferred allocation changes with the task and the available reasoning supervision.

Theorem 1 (Length-coverage trend; asymptotic form). *Consider the scalar model with the loss, dynamics, and initialization in equation 4-equation 6. W.L.O.G, let $\pi_r(e) = 1 + ((e + r - 2) \bmod N)$ for $e \in [N]$ and $r \in [M]$, with $2 \leq M \leq N$. Fix $L \geq 2$, $p = 1 - q \in (0, 1)$, $\tau > 0$, and $\delta \in (0, 1)$. Draw $\mathcal{H}$ uniformly among subsets of qNM facts, where qNM is integral, and set $\mathcal{S} = \mathcal{A} \setminus \mathcal{H}$.*

Along dimensions satisfying $M/\log(N^2M) \rightarrow \infty$, for all sufficiently large N, M , with probability at least $1 - \delta$, the reasoning population is nonempty, $\beta_{\tau,S}^$ is unique and belongs to $(1/3, 2/3)$, and*

$$|\beta_{\tau,S}^* - \Theta_{N,M}(L, 1-q)| \leq \mathfrak{R}, \quad \mathfrak{R} = O_{L,q,\tau} \left(\frac{1}{M} + \sqrt{\frac{\log(1/\delta)}{NM}} + \frac{1}{NM} \right). \quad (8)$$

For fixed N, M , the center satisfies

$$\partial_L \Theta_{N,M}(L, p) > 0, \quad \partial_q \Theta_{N,M}(L, 1-q) < 0, \quad (9)$$

where L is continuously interpolated over $[2, \infty)$.

We refer the complete proof to Section A. The theorem predicts a larger leading optimal atomic weight for longer reasoning chains and a smaller weight when a larger fraction of facts is excluded from reasoning traces, which is consistent with Figure 1.

4 EXPERIMENTS ON NATURAL LANGUAGE

Balancing general math text and reasoning data

We pretrain Olmo-3 150M on mixtures of FineMath, a general math web-text corpus, and the OpenMathInstruct-2 subset of MathLake, a mathematical reasoning dataset. We sweep the MathLake fraction from 0% to 100% in 10% steps. The total budget is fixed at 10B tokens, so higher MathLake fractions also mean more repetition of its 277M tokens (up to ~ 36 epochs). Prior work either varies composition and token count together (Zhao et al., 2025), uses a single reasoning fraction (Akter et al., 2026), or tests only small domain fractions (Baek et al., 2026); none includes both pure-source endpoints, so none shows that a mixture beats either source alone.

Table 1 shows that both extremes are suboptimal. Pure FineMath reaches a GSM8K CE of 1.785, while pure MathLake degrades to 7.513. Although its training loss falls to 0.2, GSM8K CE rises throughout training, consistent with overfitting to a heavily repeated corpus. A small share of reasoning data helps significantly, with CE dropping to 1.613 at 10% MathLake and reaching its minimum of 1.608 at 20%. Beyond this point, CE rises monotonically, returning to the pure FineMath level at 50%. We therefore run subsequent experiments at 10%, 20%, and 30% MathLake, covering the optimum and the start of the degradation regime.

Table 1: Mixture-ratio sweep between FineMath and MathLake (Olmo-3 150M, 10B tokens). Zero-shot GSM8K test CE and PPL over the solution sequence. Best in bold.

FM (%)	ML (%)	CE $\downarrow$	PPL $\downarrow$
100	0	1.785	5.96
90	10	1.613	5.02
80	20	1.608	4.99
70	30	1.674	5.33
60	40	1.689	5.41
50	50	1.768	5.86
40	60	1.817	6.16
30	70	2.035	7.66
20	80	2.148	8.57
10	90	2.402	11.04
0	100	7.513	1831.40

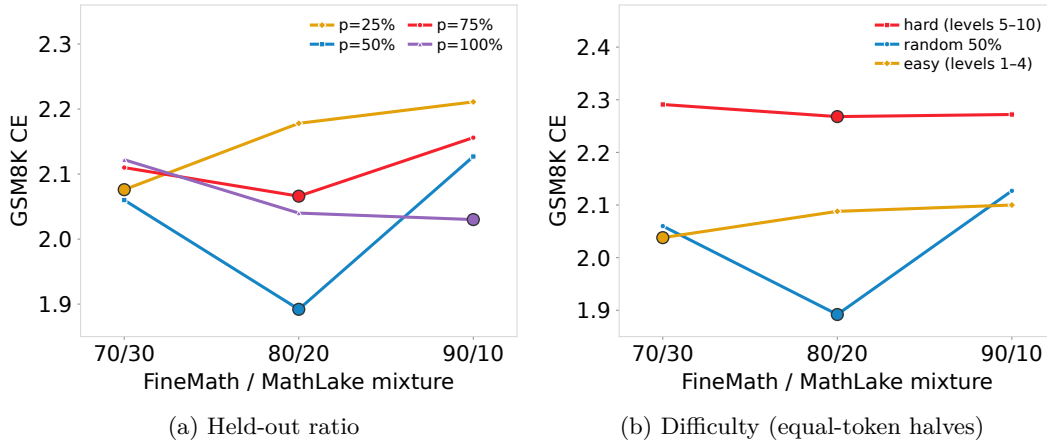


Figure 2: GSM8K CE at step 5000. (a) Varying the fraction p of the MathLake pool available for training, at three FineMath/MathLake ratios. (b) Training on the easy or hard half of MathLake versus a random 50% subset.

Quantity and difficulty of reasoning data At each of the mixture ratios above, we vary two properties of the reasoning data.

Held-out ratio. We control the fraction $p \in \{25\%, 50\%, 75\%, 100\%\}$ of the MathLake pool available for training and withhold the rest. Since the mixture ratio and total budget are fixed, every run sees the same number of reasoning tokens. To make the held-out ratio a monotone intervention, the pools are nested. We shuffle MathLake once at the document level and take each pool as a prefix of the next, where the 25% pool is a subset of the 50% pool, which is a subset of the 75% pool. Thus, moving from a smaller to a larger p only adds unseen documents and never swaps one sample for another.

Table 2: Timing of reasoning data. Each run trains Olmo-3 150M for 10k steps with MathLake making up 20% of all tokens; the first K steps use pure FineMath, so the MathLake share while mixing rises with K . Zero-shot GSM8K test CE and PPL over the solution sequence at the final checkpoint. Best in bold.

Pure FM steps (K)	ML share while mixing	CE ↓	PPL ↓
0	20.0%	1.744	5.72
1k	22.2%	1.749	5.75
2k	25.0%	1.754	5.78
3k	28.6%	1.767	5.85
4k	33.3%	1.771	5.88
6k	50.0%	1.853	6.38
8k	100.0%	2.411	11.14

Difficulty. We use MathLake’s per-problem difficulty scores (1–10) as a proxy for reasoning length and complexity. The scores closely track solution length (258 tokens at level 1 vs. 680 at level 9), so splitting by level alone would also split by token volume. We instead cut the corpus at its token-weighted median, which falls between levels 4 and 5, and cap both halves at 133.9M tokens, so easy and hard runs see identical reasoning-token budgets at identical repetition rates. A random 50% pool (138.7M tokens) serves as the control.

Results. As shown by Figure 2a, the optimal mixing ratio increases with p , therefore decreases with held-out ratio. In Figure 2b, the optimal mixing ratio increases with difficulty. The results are consistent with Theorem 1.

When to introduce reasoning data Finally, we fix the total reasoning budget and vary when it enters training. Each run lasts 10k steps with MathLake making up 20% of all training tokens, but the first K steps use FineMath only. The full reasoning budget must then fit into the remaining $10k - K$ steps, so MathLake’s share of the post-switch mixture is 20%, 25%, 33%, 50%, and 100% for $K = 0, 2k, 4k, 6k, 8k$. We implement this by resuming each arm from the step- K checkpoint of a single pure-FineMath run, carrying over the step count and learning-rate schedule.

Delaying reasoning data brings no benefit (Table 2): GSM8K CE increases monotonically with K . Since starting early is never worse, we recommend mixing in reasoning data from the beginning of training, consistent with Baek et al. (2026), who find that domain data is best introduced early.

5 CONCLUSION

We studied a fundamental question in reasoning-oriented pretraining: how should a fixed training budget be divided between atomic facts and reasoning traces? By separating factual coverage from exposure to multi-step composition, we showed that the optimal allocation depends systematically on reasoning-chain length and reasoning-trace coverage. Perhaps surprisingly, longer reasoning chains can favor greater allocation to atomic facts, as richer compositional training comes at the cost of broader factual coverage. Our theoretical analysis explains this effect in a simplified linear setting, and experiments with full Transformers and realistic mathematical data demonstrate that the same qualitative trends persist beyond the controlled regime. Together, our results provide a principled perspective on how the structure and coverage of reasoning data should shape pretraining mixtures.

Limitations. Our controlled setting abstracts away heterogeneous data sources and varying data quality in large-scale pretraining. We also focus on pretraining and do not study how different data mixtures affect subsequent SFT and RL, which we leave for future work.

AI USE STATEMENT

We have not used generative AI tools for tasks with required disclosure. Additionally, we used generative AI tools for polishing the paper. We have reviewed all AI-assisted work. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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A PROOF OF THE LENGTH-COVERAGE THEOREM

This appendix proves Theorem 1 through its finite-dimensional version, Theorem 2. We keep the scalar outputs, shared-prefix populations, and joint training of the active W, B, C parameters from Section 3.3. The explicit constants are collected at the end in Appendix A.9.

Proof outline. There are three ingredients. Lemmas 2–4 identify the deterministic center and prove its two trend directions. Lemma 15 bounds the difference between the random-mask cubic coefficients and their deterministic references. Lemma 19 controls the remaining finite-time error for the actual factor dynamics. Lemma 20 combines these two errors to locate the exact optimal mixture. Appendix A.7 then completes the formal theorem, and Appendix A.8 deduces the main-text asymptotic form.

A.1 SETUP AND ANALYTIC CENTER

Let $[K] = \{1, \dots, K\}$, $2 \leq M \leq N$, and

$$\pi_r(e) = 1 + ((e + r - 2) \bmod N), \quad \mathcal{A} = [N] \times [M], \quad D = NM. \quad (10)$$

This is $(i+j) \bmod N$ with zero-based entity and relation indices. A chain $\chi = (r_1, \dots, r_L, e_0)$ has $e_\ell = \pi_{r_\ell}(e_{\ell-1})$. Put $y_{r,e} = \zeta_{\pi_r(e)}$, with the one-based scalar code in equation 3; thus $|y_{r,e}| \leq 1$. The fact order is (e, r) and the corresponding table entry is $b_{r,e}$.

Draw H uniformly among subsets of $\mathcal{A}$ of size qD , where $0 < q < 1$ and qD is integral. Put $p = 1 - q$ and $S = \mathcal{A} \setminus H$. The mask is sampled once and fixed throughout training. The chain populations are

$$\begin{aligned} \mathcal{D}_{\text{at}} &= \mathcal{D}_{\text{full}} = \text{Unif}([M]^L \times [N]), \\ \mathcal{C}_L(S) &= \{\chi : (e_{\ell-1}, r_\ell) \in S \text{ for every } \ell \in [L]\}, \\ \mathcal{D}_{\text{rea}}(S, L) &= \text{Unif}(\mathcal{C}_L(S)). \end{aligned}$$

The first population samples all relation labels and the initial entity independently. Both sources have input $[r_1, \dots, r_L, e_0]$. Atomic loss supervises only e_1 , while reasoning loss supervises the whole teacher-forced trace. Uniform valid-chain sampling is not replaced by uniform local choices among retained outgoing relations. The event constructed below ensures $\mathcal{C}_L(S) \neq \emptyset$.

In the content encoding, a relation has vector $(u_r^\top, 0_N^\top)^\top$ and an entity has vector $(0_M^\top, v_e^\top)^\top$, where u_r, v_e are standard basis vectors. All embeddings are fixed. The loss, dynamics, and held-role evaluation are those of equation 4–equation 7. The main theorem uses $B(0) = 0$. To state the finite-dimensional extension, we also allow

$$w_r(0) = 1, \quad b_{r,e}(0) = \sigma Z_{r,e}, \quad |Z_{r,e}| \leq 1, \quad C(0) = E_{11}, \quad \sigma \geq 0. \quad (11)$$

The directions Z are fixed across mixture comparisons. Setting $\sigma = 0$ recovers equation 6; for either initialization, all active entries are jointly trained.

Expectation notation. We write $\mathbb{E}_S$ and Pr_S for the uniform valid-chain law conditional on a fixed retained set S , and $\mathbb{E}_{H,\ell}$ for the full-world law conditional on a held actual transition at role ℓ . Probabilities and expectations over the random mask itself are identified explicitly in the concentration arguments. Chain expectations do not resample the mask.

Reference moments. The scalar-code variance and source-mean energy are

$$\begin{aligned} \nu &= \frac{1}{N} \sum_{e=1}^N \zeta_e^2 = \frac{N^2 - 1}{3N^2}, \\ s &= \frac{1}{N} \sum_{e=1}^N \left(\frac{1}{M} \sum_{r=1}^M y_{r,e} \right)^2 = \nu - \frac{(M^2 - 1)(2N - M)}{3MN^2}, \\ \kappa_{N,M} &= (s/\nu)^2. \end{aligned} \quad (12)$$

Equivalently,

$$\kappa_{N,M} = \left[1 - \frac{(M^2 - 1)(2N - M)}{M(N^2 - 1)} \right]^2.$$

Lemma 2 proves this identity. Define

$$\begin{aligned} A_o &= \nu^2 + (L - 1)s^2, & B_o &= \frac{\nu^2}{pL} + \frac{2(L - 1)s^2}{L}, & C_o &= \frac{(L - 1)s^2}{L^2}, \\ u &= B_o/A_o, & v &= C_o/A_o, \\ R_o(\beta) &= (1 - \beta)[\beta^2 + u\beta(1 - \beta) + v(1 - \beta)^2]. \end{aligned} \quad (13)$$

For comparison, write $\mathfrak{b}(z) = [2 - z + \sqrt{z^2 - z + 1}]^{-1}$. Thus $u = [1/(pL) + 2\kappa_{N,M}(L - 1)/L]/[1 + \kappa_{N,M}(L - 1)]$ and $v = [\kappa_{N,M}(L - 1)/L^2]/[1 + \kappa_{N,M}(L - 1)]$. Fix an auxiliary radius $\rho > 0$ independently of the random mask. All bounds below refer to the same choice of ρ ; it is suppressed in the main-text notation $\mathfrak{R}$. For the asymptotic statement, any one fixed positive choice, such as $\rho = 0.1$, may be used.

The analytic center used in both theorem statements is

$$\Theta_{N,M}(L, p) = \frac{1 - u + \sqrt{1 - u + u^2 - 3v}}{2 - u + \sqrt{1 - u + u^2 - 3v}}. \quad (14)$$

It depends only on the world, length, and coverage, not on the realized mask or a trained optimizer. Its characterization as the unique maximizer of R_o and its strict monotonicity are proved in Appendix A.4.

A.2 FINITE-DIMENSIONAL STATEMENT

Theorem 2 (Quantitative length-coverage trend). *Consider the setup above and joint scalar W, B, C training from equation 11. Let $pD \geq 2L$, $0 < \delta < 1$, $\rho > 0$, $\tau > 0$, and $\sigma \geq 0$. Define the deterministic constants and the radius $\mathfrak{R}$ by Appendix A.9. Assume the admissibility conditions equation 64. Then, with probability at least $1 - \delta$ over the fixed-cardinality mask, the reasoning population is nonempty, the exact optimizer is unique in $(1/3, 2/3)$, and*

$$|\beta_{\tau,S}^* - \Theta_{N,M}(L, p)| \leq \mathfrak{R} = \frac{d_u(1 - \Theta)(3\Theta - 1) + 3d_v(1 - \Theta)^2 + T_1}{\mu_o - d_2}, \quad \Theta = \Theta_{N,M}(L, p). \quad (15)$$

The center satisfies equation 9.

All conditions are deterministic functions of $N, M, L, p, \delta, \rho, \tau, \sigma$.

Recurring proof quantities.

P_S	Exact scalar cubic for a realized retained set S .
R_o	Deterministic normalized reference cubic, with maximizer Θ .
$\mathcal{G}$	Event controlling every retained incoming and outgoing degree.
$\mathcal{E}$	Common probability event controlling the needed scalar moments.
d_u, d_v	Bounds on the two normalized cubic-coefficient errors.
T_1, T_2	Bounds on the first two mixture derivatives of the finite-time remainder.

The functions P_S and R_o are not training objectives. They describe the leading held-evaluation improvement; the exact criterion remains $\mathcal{E}_H(\theta_\beta(\tau))$.

A.3 FROM THE FULL TRANSFORMER TO SCALAR LINEAR ATTENTION

Supported parameter blocks. The active blocks in Definition 6 are

$$W = [w^\top \ 0_N^\top], \quad B = \begin{pmatrix} 0_{M \times M} & b \\ 0_{N \times M} & 0_{N \times N} \end{pmatrix}, \quad b \in \mathbb{R}^{M \times N}, \quad w \in \mathbb{R}^M, \quad (16)$$

with the positional support

$$\tilde{C}_{uv} = \begin{cases} C_{j\ell}, & u = j, v = L + \ell, \quad j, \ell \in [L], \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

Lemma 1 (Exact scalar representation). *For the supported blocks equation 16–equation 17, every supervised prediction has the exact scalar form*

$$\begin{aligned} f_\ell(\chi) &= \sum_{j=1}^L C_{j\ell} (W \mathbf{c}_j) (\mathbf{c}_j^\top B \mathbf{c}_t) \\ &= \sum_{j=1}^L C_{j\ell} w_{r_j} b_{r_j, e_{\ell-1}} = \sum_{j=1}^L C_{j\ell} G_{r_j, e_{\ell-1}}, \quad G_{r,e} := w_r b_{r,e}. \end{aligned} \quad (18)$$

Proof. At query position $t = L + \ell$, the only allowed keys are relation positions $j \in [L]$. The typed one-hot vectors give $W \mathbf{c}_j = w_{r_j}$ and $\mathbf{c}_j^\top B \mathbf{c}_t = b_{r_j, e_{\ell-1}}$. Expanding the query column of equation 2 therefore yields equation 18. Both sources use the same first query position $L + 1$ and the entire column $C_{:1}$. $\square$

A.4 THE ANALYTIC CENTER AND ITS TREND DIRECTIONS

We first analyze the deterministic reference cubic. This part is independent of the random mask and the training trajectory.

Lemma 2 (Cyclic scalar correlations). *For the consecutive modular shifts in equation 10, equation 12 holds. Moreover $0 \leq s \leq \nu$ and, for two shifts whose difference is $d \in \{0, \dots, N-1\}$,*

$$\frac{1}{N} \sum_e y_{r,e} y_{r+d,e} = \nu - \frac{2d(N-d)}{N^2}. \quad (19)$$

Proof. The codes have mean zero and second moment $(N^2 - 1)/(3N^2)$ by summing integers and their squares. Shifting a code by d changes it by $2d/N$ on $N - d$ entries and by $-2(N - d)/N$ on d entries. The mean squared change is $4d(N - d)/N^2$. Since both shifted rows have second moment ν , polarization gives equation 19. Averaging over all ordered pairs of shifts gives

$$s = \nu - \frac{4}{M^2 N^2} \sum_{d=1}^{M-1} (M - d) d (N - d).$$

Using the formulas for the first three integer power sums,

$$\sum_{d=1}^{M-1} (M - d) d (N - d) = \frac{M(M^2 - 1)(2N - M)}{12},$$

which proves equation 12. Nonnegativity follows from the mean-square definition of s . $\square$

Lemma 3 (Maximizer and curvature of the reference cubic). *Fix $0 \leq \kappa \leq 1$, $L \geq 2$, and $p \in (0, 1)$ in the coefficient formulas equation 13. The cubic $R_\circ$ has a unique maximizer $\Theta \in (1/3, 2/3)$, given by equation 14. Moreover,*

$$\inf_{[0, 1/3]} R'_\circ \geq m_\circ, \quad \inf_{[2/3, 1]} (-R'_\circ) \geq m_\circ, \quad \inf_{[1/3, 2/3]} (-R''_\circ) = \mu_\circ > 0. \quad (20)$$

where $m_\circ, \mu_\circ$ are the positive explicit margins in equation 58.

Proof. Location of the maximizer. The formulas imply $u > 3v$ and $0 \leq v \leq (L-1)/L^3 \leq 1/8$. Direct differentiation gives

$$R'_\circ(0) = u - 3v > 0, \quad R'_\circ(1/3) = \frac{1 - 4v}{3} > 0, \quad R'_\circ(2/3) = -\frac{u + v}{3} < 0, \quad R'_\circ(1) = -1.$$

The quadratic derivative has precisely one zero in $(0, 1)$, since it has opposite endpoint signs. With odds $z = \beta/(1 - \beta)$, its stationary equation is

$$-z^2 + (2 - 2u)z + u - 3v = 0.$$

Its positive root is $z = 1 - u + \sqrt{1 - u + u^2 - 3v}$, giving equation 14. The signs at $1/3, 2/3$ locate the root in that interval.

Uniform slope and curvature margins. The negative second derivative is affine and has endpoint values $2(u - 2v)$ and $2(1 - v)$ on $[1/3, 2/3]$, which proves the curvature formula. On the left exterior, if R'_o is concave, its minimum occurs at an endpoint. If it is convex, its derivative R''_o is increasing and is negative at $1/3$, so R'_o is decreasing throughout that exterior. The right exterior is treated by the same endpoint argument, or by monotonicity if R'_o is concave. This proves all of equation 20. $\square$

Lemma 4 (Length and held-fraction trends). *With the world dimensions N, M fixed, the center in equation 14 satisfies*

$$\partial_L \Theta_{N,M}(L, p) > 0, \quad \partial_q \Theta_{N,M}(L, 1 - q) < 0, \quad L \geq 2, \quad 0 < p < 1.$$

Here L is continuously interpolated and $\kappa_{N,M}$ is held fixed.

Proof. Sensitivity to the two coefficients. At the stationary point,

$$\partial_u R'_o(\Theta) = (1 - \Theta)(1 - 3\Theta) < 0, \quad \partial_v R'_o(\Theta) = -3(1 - \Theta)^2 < 0.$$

Since $R''_o(\Theta) < 0$, implicit differentiation gives $\Theta_u < 0$ and $\Theta_v < 0$.

Dependence of the coefficients on length and coverage. Put $c_p = 1/p \geq 1$ and $A = 1 + \kappa(L - 1)$. Then

$$u_p = -\frac{1}{p^2 LA} < 0, \quad v_p = 0, \tag{21}$$

$$u_L = -\frac{2\kappa^2 L(L - 2) + 2\kappa c_p L + (1 - \kappa)(c_p - 2\kappa)}{L^2 A^2} < 0, \tag{22}$$

$$v_L = \frac{\kappa[(2 - L)A - \kappa L(L - 1)]}{L^3 A^2} \leq 0. \tag{23}$$

The numerator negated in equation 22 is at least $1 + \kappa + 2\kappa^2 > 0$, using $L \geq 2, c_p \geq 1$. Therefore $\Theta_p > 0$, $\Theta_q = -\Theta_p < 0$, and $\Theta_L > 0$. $\square$

A.5 RANDOM-MASK CONTROL OF THE SCALAR CUBIC

The objective of this subsection is Lemma 15: with high probability, the three coefficients of P_S are close to their deterministic references. We first express the moments as unnormalized path sums, then separate their expectation bias from their fluctuations. The degree event is used only to control sensitivity to a changed mask edge.

A.5.1 FIXED-MASK MOMENTS AND HELD-ROLE CONDITIONING

Write $a_{\ell j} = (e_{\ell-1}, r_j)$ and $a_\ell = a_{\ell\ell}$. For a fixed mask with a nonempty reasoning population, define

$$\omega_a = \frac{D}{L} \Pr_S(a_1 = a), \quad \nu_H = \frac{1}{qD} \sum_{a \in H} y_a^2, \quad \mu_e = \frac{1}{M} \sum_r y_{r,e}.$$

The four signed moments needed below are

$$\begin{aligned} K_{j\ell} &= \mathbb{E}_S[y_{a_{\ell j}} y_{a_\ell}], & U_{j\ell} &= \mathbb{E}_S[\omega_{a_{\ell j}} y_{a_{\ell j}} y_{a_\ell}], \\ H_{j\ell} &= \mathbb{E}_{H,\ell}[y_{a_{\ell j}} y_{a_\ell}], & Z_{j\ell} &= \mathbb{E}_{H,\ell}[\omega_{a_{\ell j}} y_{a_{\ell j}} y_{a_\ell}]. \end{aligned} \tag{24}$$

Lemma 5 (Held-role conditioning and evaluation moments). *At each role under $\mathcal{D}_{\text{full}}$, the current state $e_{\ell-1}$ is uniform and independent of the complete relation tuple. Consequently a_ℓ is uniform on the D facts and $\Pr_{\text{full}}(a_\ell \in H) = q$. For the moments in equation 24,*

$$H_{\ell\ell} = \nu_H, \quad Z_{\ell\ell} = 0, \quad H_{j\ell} = H_O, \quad Z_{j\ell} = Z_O \quad (j \neq \ell),$$

where

$$H_O = \frac{1}{qD} \sum_{(e,r) \in H} \mu_e y_{r,e}, \quad (25)$$

$$Z_O = \frac{1}{qDM} \sum_e \left(\sum_{r:(e,r) \in H} y_{r,e} \right) \left(\sum_t \omega_{t,e} y_{t,e} \right). \quad (26)$$

Proof. Condition on the entire relation tuple. The map $e_0 \mapsto e_{\ell-1}$ is a composition of permutations, hence a permutation. Uniformity of e_0 makes the conditional law of $e_{\ell-1}$ uniform, independently of the tuple.

Conditioning further on a specified actual pair $(e, r) \in H$ leaves every relation label at a different position independent and uniform. Averaging its scalar code gives μ_e , which proves equation 25. Averaging instead its exposure-weighted code gives $M^{-1} \sum_t \omega_{t,e} y_{t,e}$, yielding equation 26 after averaging the held pair. For $j = \ell$, the first identity is the definition of ν_H , and the second follows from $\omega_{a_\ell} = 0$ for a held actual pair. None of these independence statements is asserted for the masked reasoning population. $\square$

Define the fixed-mask cubic and its three coefficients by

$$\begin{aligned} P_S(\beta) &= \frac{1-\beta}{L-1} \sum_{\ell=2}^L \sum_{j=1}^L [\beta K_{j\ell} + (1-\beta)U_{j\ell}] [\beta H_{j\ell} + (1-\beta)Z_{j\ell}] \\ &= (1-\beta)[A_S \beta^2 + B_S \beta(1-\beta) + C_S(1-\beta)^2]. \end{aligned} \quad (27)$$

The factors in this expression are separately evaluated training and evaluation moments. Their product does not factor an expectation of dependent variables within one chain. Lemma 17 will identify P_S with the first nonzero learning coefficient of the raw dynamics.

A.5.2 A DEGREE EVENT AND DETERMINISTIC PATH BOUNDS

Put $d_\pm = pM(1 \pm h)$, with h defined in equation 51, and let

$$\mathcal{G} = \{d_- \leq d_{\text{in}}(e), d_{\text{out}}(e) \leq d_+ \text{ for all } e \in [N]\}.$$

Lemma 6 (Degree control and path counts). *For a uniform retained set of size pD and $0 < h < 1$,*

$$\mathbb{P}(\mathcal{G}^c) \leq 4Ne^{-pMh^2/3} = \delta_g. \quad (28)$$

On $\mathcal{G}$, the reasoning population is nonempty, its size Z_S satisfies

$$pDd_-^{L-1} \leq Z_S \leq pDd_+^{L-1}, \quad Z_S \geq Nd_-^L. \quad (29)$$

Every candidate marginal also satisfies

$$D \max_{\ell,j,a} \Pr_S(a_{\ell j} = a) \leq \frac{Dd_+^{L-1}}{Nd_-^L} = J_+. \quad (30)$$

Repeated states, relations, and edges are allowed in these counts.

Proof. Probability of the degree event. Each degree is hypergeometric with mean pM : a source has M outgoing facts, and a destination has M incoming facts because all relations are permutations. The exponential-moment bound

$$\mathbb{E}e^{tX} \leq (1 - p + pe^t)^M \leq \exp(pM(e^t - 1))$$

for sampling without replacement gives, for $0 < h < 1$,

$$\mathbb{P}(X \geq (1+h)pM) \leq e^{-pMh^2/3}, \quad \mathbb{P}(X \leq (1-h)pM) \leq e^{-pMh^2/2}.$$

For completeness, the first inequality on exponential moments follows by averaging two unequal nonnegative entries in the elementary symmetric polynomial while keeping their sum fixed; this increases the symmetric mean, so it is at most the corresponding power of the arithmetic mean. Exponential Markov with $t = \log(1+h)$ and $t = \log(1-h)$ gives the tails. Integrating $\log(1+x) \geq x - x^2/2$ and $-\log(1-x) \geq x$ supplies the constants. A union bound gives equation 28. The event ensures a nonempty reasoning population, because every integer outgoing degree is at least one.

Counting paths and candidate occurrences. There are between d_-^k and d_+^k retained length- k paths from a fixed starting state to a free endpoint. The same bounds hold for paths ending at a fixed state with a free starting state, using the incoming degrees. Multiplying a retained first-edge choice by its continuation count proves equation 29. The number of valid chains with any fixed candidate pair $a_{\ell j} = (e, r)$ is at most d_+^{L-1} . To verify this, write A_r for the retained partial-permutation adjacency matrix of relation r and $A = \sum_r A_r$, with source states indexing rows. Row and column sums of A are at most d_+ , while those of A_r are at most one. If $j < \ell$, its prefix count is

$$\mathbf{1}^\top A^{j-1} A_r A^{\ell-1-j} v_e \leq d_+^{\ell-2},$$

and its suffix count is at most $d_+^{L-\ell+1}$. If $j \geq \ell$, the prefix count is at most $d_+^{\ell-1}$ and the constrained suffix count is

$$v_e^\top A^{j-\ell} A_r A^{L-j} \mathbf{1} \leq d_+^{L-\ell}.$$

In both cases the exponents add to $L-1$. Division by $Z_S \geq Nd_+^L$ gives equation 30. These are path-count statements with repeated states and edges allowed. $\square$

A.5.3 UNNORMALIZED PATH SUMS

For a mask-independent weight f with $|f| \leq 1$, let

$$F_0(S) = \mathbb{E}_{\text{full}} I_S(\chi) = \frac{Z_S}{NM^L}, \quad F_f(S) = \mathbb{E}_{\text{full}} [I_S(\chi) f(\chi)],$$

where $I_S(\chi)$ indicates that every actual edge of χ is retained. For $K_{j\ell}$, the weight is $f = y_{a_{\ell j}} y_{a_\ell}$.

For $F_{j\ell}^{\text{pair}}$, sample an original full-world path χ and an auxiliary length- L path starting at $e_{\ell-1}$. Force the auxiliary first relation to be r_j , and sample its remaining $L-1$ relations independently and uniformly. Average the product of both retention indicators and the weight $y_{a_{\ell j}} y_{a_\ell}$. For F^{mix} , sample a full-world path starting at a uniform e and an independent relation r_0 , and average

$$I_S(\chi) \mathbf{1}\{(e, r_0) \in H\} y_{r_0, e} y_{r_1, e}.$$

These raw functions exist even when the mask has no valid chains. The auxiliary path is used only to express a statistic; it is not additional training data.

Lemma 7 (Raw-to-conditional moment identities). *Whenever $F_0(S) > 0$, the conditional moments satisfy*

$$K_{j\ell} = \frac{F_f}{F_0}, \quad U_{j\ell} = \frac{F_{j\ell}^{\text{pair}}}{LF_0^2}. \quad (31)$$

$$Z_O = \frac{F^{\text{mix}}}{qLF_0}. \quad (32)$$

Proof. The identity for $K_{j\ell}$ is the definition of conditioning on $I_S(\chi) = 1$. For a fixed first fact a , write $h_S(a)$ for the retention probability of a full path with that first fact forced. Since the unconditioned first fact has probability D^{-1} ,

$$\Pr_S(a_1 = a) = \frac{h_S(a)}{DF_0(S)}, \quad \omega_a = \frac{h_S(a)}{LF_0(S)}.$$

Substitute this expression for $\omega_{a_{\ell j}}$ into the definition of $U_{j\ell}$. The first conditioning contributes F_0^{-1} , and the exposure contributes a second F_0^{-1} ; the remaining numerator is exactly $F_{j\ell}^{\text{pair}}$.

Similarly,

$$F^{\text{mix}} = \frac{1}{DM} \sum_e \left(\sum_{r:(e,r) \in H} y_{r,e} \right) \left(\sum_t h_S(e,t) y_{t,e} \right).$$

Substitution into equation 26 proves the qLF_0 normalization. $\square$

Lemma 8 (Sensitivity of the raw path statistics). *On $\mathcal{G}$, use the metric $\text{dist}(S, S') = |S \setminus S'|$ (number of exchanged facts). The following functions are Lipschitz, with the indicated constants from equation 52:*

raw function	Lipschitz constant
F_0/p^L and F_f/p^L	c_0
$F_{\ell\ell}^{\text{pair}}/p^{2L-1}$	c_D
$F_{j\ell}^{\text{pair}}/p^{2L}$ ($j \neq \ell$)	c_O
$F^{\text{mix}}/(qp^L)$	c_Z

The assertion holds between any two masks in $\mathcal{G}$, without assuming that an exchange path between them remains in $\mathcal{G}$.

Proof. Single paths. Consider two masks in $\mathcal{G}$ at distance k . A path whose retention indicator changes must use one of the k removed or k added facts. At a specified role, the number of retained paths through a specified edge is at most d_+^{L-1} . Union bounding the roles gives

$$|F_f(S) - F_f(S')| \leq \frac{2kLd_+^{L-1}}{NM^L}.$$

After division by p^L this is kc_0 , including for F_0 .

Coupled paths. For a pair configuration, the total number of unmasked possibilities is NM^{2L-1} . If a tagged edge is in the first path, there are at most d_+^{L-1} first paths through it and at most d_+^{L-1} second-path tails. If it is in the second path, fix that path's initial candidate (e, r) . The first-path candidate count is at most d_+^{L-1} by the preceding candidate argument; summing the second paths through the tagged edge gives at most another d_+^{L-1} . Thus in either case there are at most d_+^{2L-2} pair configurations per tagged edge and position. There are $2L$ positions for an off-diagonal pair; for $j = \ell$, the forced first edge of the second path is already the original actual edge at role ℓ , so only $2L - 1$ positions need be counted. Therefore the scaled statistics

$$F_{\ell\ell}^{\text{pair}}/p^{2L-1}, \quad F_{j\ell}^{\text{pair}}/p^{2L} \quad (j \neq \ell) \quad (33)$$

have respective Lipschitz constants c_D and c_O in equation 52. This direct count avoids a ratio-of-populations bound and its unnecessary powers of Λ_h .

A retained path and a held test edge. For F^{mix} , changes through a path edge give at most $2kLMd_+^{L-1}$ configurations; the free held-candidate relation is bounded by M choices. Changes through the held test edge fix its source and relation, leaving at most d_+^L paths per changed edge. Dividing by the total NM^{L+1} possibilities and by qp^L gives kc_Z . Thus $F^{\text{mix}}/(qp^L)$ has constant c_Z . Absolute values of scalar weights are bounded before counting, so no sign cancellation is assumed. $\square$

A.5.4 EXPECTATION BIAS FROM REPEATED FACTS

The raw statistics are centered at their unmasked moments. Their mask expectations are not exactly those moments: repeated edge requests and sampling without replacement both contribute a bias. The next three lemmas account for these effects before concentration.

Lemma 9 (Inclusion probabilities with repeated requests). *Let a configuration request $k \leq 2L$ retained edges, of which $d = k - R$ are distinct, and assume $pD \geq k$. Its retention probability is*

$$p_d = \frac{(pD)_d}{(D)_d}, \quad (x)_d = x(x-1) \cdots (x-d+1).$$

With $f_k = k(k-1)/(2pD)$,

$$1 - f_k \leq \frac{p_d}{p^k} \leq p^{-R}, \quad \mathbb{E}|p_d/p^k - 1| \leq f_k + \mathbb{E}(p^{-R} - 1). \quad (34)$$

The expectation in the second bound is over the sampled path configuration; R counts duplicates.

Proof. The lower bound follows by dividing first by p^d , bounding each factor below by $1 - j/(pD)$, and using $\prod(1 - x_j) \geq 1 - \sum x_j$ for $x_j \in [0, 1]$. The upper bound follows from $p_d \leq p^d$.

The pointwise lower and upper bounds imply $|p_d/p^k - 1| \leq f_k + (p^{-R} - 1)$. Taking the configuration expectation proves the second inequality. $\square$

Lemma 10 (Repeated edges on one path). *For one full-world modular path of length L , let R be its number of repeated actual edges after their first occurrence. With $t_p = q/p$,*

$$\mathbb{E}(p^{-R} - 1) \leq \frac{(1 + t_p L/M)^{L-1} - 1}{M} = b_{\text{self}}. \quad (35)$$

Proof. For one unmasked modular path, let I_i indicate that edge i repeats an earlier edge. A sum of any positive number of independent uniform shifts has maximum point mass at most $1/M$, because convolution cannot increase that maximum. For positions $i > j$, equality of the source states has probability at most $1/M$, and equality of the fresh relation r_i with r_j has conditional probability $1/M$. Hence

$$\mathbb{P}(I_i = 1) \leq (i-1)/M^2 \leq L/M^2, \quad \mathbb{P}(I_i = 1 \mid \text{past}) \leq L/M.$$

For indices $2 \leq i_1 < \cdots < i_k \leq L$, the unconditional bound applies to I_{i_1} , while successive conditional bounds apply to the remaining indicators. Hence

$$\mathbb{E} \left[\prod_{j=1}^k I_{i_j} \right] \leq \frac{L}{M^2} \left(\frac{L}{M} \right)^{k-1}.$$

Expanding $\prod_{i=2}^L (1 + t_p I_i)$ and summing these bounds over all subsets gives

$$\sum_{k=1}^{L-1} \binom{L-1}{k} t_p^k \frac{L}{M^2} \left(\frac{L}{M} \right)^{k-1} = \frac{(1 + t_p L/M)^{L-1} - 1}{M},$$

which is equation 35. This argument does not assert that repetitions at different positions are independent. $\square$

Lemma 11 (Expectation bias of the path statistics). *Assume $pD \geq 2L$. Let $\mathbb{E}_{\text{mask}}$ denote expectation over the uniform fixed-cardinality mask. For any fixed $|f| \leq 1$,*

$$|\mathbb{E}_{\text{mask}}(F_f/p^L) - \mathbb{E}_{\text{full}} f| \leq b_{\text{self}} + f_L.$$

For single-path, diagonal-pair, off-diagonal-pair, and mixed statistics, respectively, the expectation-bias bounds are

$$b_{\text{self}} + f_L, \quad b_D + f_{2L-1}, \quad b_O + f_{2L}, \quad b_Z. \quad (36)$$

In this display, the first entry is the single-path bound just stated. The reference mean is ν for diagonal weights and s for off-diagonal and mixed weights. The constants b_D, b_O, b_Z are defined in equation 54.

Proof. Single-path terms. Lemmas 9 and 10 give the first bound, since multiplying by a weight of absolute value at most one does not increase the absolute retention-probability error.

Pair terms, including the forced first edge. For a pair statistic, expose the original path first. After the forced first edge of the second path, each remaining relation is fresh, so its conditional probability of repeating any already exposed edge is at most $2L/M$. These $L - 1$ positions contribute at most the factor F_*^{col} . In the diagonal case the forced first edge is a known duplicate removed from the nominal count $2L - 1$, giving b_D . In an off-diagonal case the first forced candidate is itself a duplicate with probability at most L/M : if it matches the original edge at position j , a nonempty return segment has probability at most $1/M$; if it matches any other original position, the two independent relation labels must coincide, again with probability $1/M$. If I denotes this event and R_1 the original path's duplicates, then

$$\mathbb{E}[p^{-R_1} I] \leq \mathbb{P}(I) + \mathbb{E}(p^{-R_1} - 1) \leq L/M + b_{\text{self}}.$$

Thus the expected inflation before the fresh tail is at most $1 + b_{\text{self}} + t_p(L/M + b_{\text{self}}) = 1 + b_{\text{self}}/p + t_p L/M$, giving b_O .

Held/retained conflicts. For F^{mix} , the independently chosen held edge can equal a path edge. Equality with the first edge has probability $1/M$; with a later edge it requires a return to the starting source and an independent relation equality, with probability at most $1/M^2$. The conflict probability is at most $1/M + (L - 1)/M^2$. When there is no conflict and the path has d distinct edges, its retention/held probability is

$$p_d \frac{qD}{D - d} = \frac{qp_d}{1 - d/D}.$$

The possible downward deviation after division by qp^L is at most f_L ; the upward deviation is at most $[(p^{-R} - 1) + L/D]/(1 - L/D)$. Add the conflict bound and use equation 35 to obtain b_Z .

The unmasked weighted reference means are ν for a diagonal K or pair statistic and s for an off-diagonal one, by permutation conditioning. The unmasked mixed reference is also s . Applying Lemma 9 to these cases gives equation 36. Only distinct-edge probabilities were multiplied; repeated edges were accounted for explicitly. $\square$

A.5.5 CONCENTRATION WITHOUT CONDITIONING ON THE DEGREE EVENT

Lemma 8 controls sensitivity only on $\mathcal{G}$. We therefore extend each raw function to the complete fixed-size mask space before applying concentration. We never assume that the mask remains uniform after conditioning on $\mathcal{G}$.

Lemma 12 (Concentration on a fixed-cardinality slice). *Let F be a real function on masks of retained size pD that changes by at most c per exchanged fact. Assume $c > 0$. If $k = \min\{p, q\}D$, then for every $t > 0$,*

$$\mathbb{P}(|F - \mathbb{E}F| > t) \leq 2 \exp\left(-\frac{2t^2}{kc^2}\right), \quad k = \min\{p, q\}D. \quad (37)$$

Here probability and expectation are both over the random mask.

Proof. To prove it, reveal the smaller of the retained and held subsets one element at a time. Given the previous reveals, couple two different values of the next reveal by exchanging those elements in the remaining uniformly sampled subset. Final masks then differ by at most one exchange. Thus the conditional range of each Doob increment has length at most c . A centered variable in an interval of length c has log moment-generating function at most $t^2 c^2 / 8$: its tilted variance is at most $c^2 / 4$, and twice integrating the second derivative proves the claim. Iterating the conditional exponential bound over k reveals and optimizing exponential Markov proves equation 37. $\square$

Lemma 13 (Extension from the degree event). *Set $U_* = 1/(qp^{2L})$, and suppose a raw function F is c -Lipschitz on $\mathcal{G}$ with $|F| \leq U_*$ on the entire slice. Assume the degree bound in Lemma 6, so $\mathcal{G}$ is nonempty. There is a globally c -Lipschitz extension $\tilde{F}$ agreeing with F on $\mathcal{G}$, with $|\tilde{F}| \leq U_*$, and*

$$|\mathbb{E}\tilde{F} - \mathbb{E}F| \leq 2U_*\mathbb{P}(\mathcal{G}^c) \leq 2U_*\delta_g = e_{\text{ext}}. \quad (38)$$

For the raw statistics here, take $U_* = 1/(qp^{2L})$; then the right-hand side is $e_{\text{ext}} = \delta/(8D)$.

Proof. Define the extension on the whole slice by

$$\tilde{F}(S) = \inf_{S' \in \mathcal{G}} \{F(S') + c \text{dist}(S, S')\},$$

and clip to $[-U_*, U_*]$ with $U_* = 1/(qp^{2L})$. The triangle inequality gives the same Lipschitz constant. For $S \in \mathcal{G}$, the Lipschitz property on $\mathcal{G}$ makes every term in the infimum at least $F(S)$, and $S' = S$ attains equality. Clipping preserves both properties; the raw functions themselves have this global absolute bound. The event $\mathcal{G}$ is nonempty by equation 28. Since the two functions agree on $\mathcal{G}$ and are bounded by U_* elsewhere, equation 38 follows. $\square$

Lemma 14 (One simultaneous scalar-moment event). *Assume $pD \geq 2L$ and $0 < h < 1$, with the constants from equation 51–equation 55. There is an event $\mathcal{E} \subseteq \mathcal{G}$ on which the following bounds hold simultaneously for all $\ell \in \{2, \dots, L\}$ and $j \in [L]$:*

$$\begin{aligned} |F_0/p^L - 1| &\leq e_0, & |F_f/p^L - \mathbb{E}_{\text{full}}f| &\leq e_0, \\ |F_{\ell\ell}^{\text{pair}}/p^{2L-1} - \nu| &\leq e_D, & |F_{j\ell}^{\text{pair}}/p^{2L} - s| &\leq e_O \quad (j \neq \ell), \\ |F^{\text{mix}}/(qp^L) - s| &\leq e_Z, & |\nu_H - \nu|, |H_O - s| &\leq e_H. \end{aligned}$$

Here $f = y_{a_{\ell j}} y_{a_{\ell}}$; only this finite set of weights is included. The event has probability at least

$$1 - \delta_g - \delta/2 \geq 1 - \delta. \quad (39)$$

Proof. For each extended raw function, apply Lemma 12 with $t = cv_\delta$. The definition of v_δ gives failure probability at most $\delta/(2B_*)$. On $\mathcal{G}$, replace $\tilde{F}$ by F , use Lemma 13 for the mean discrepancy, and add the appropriate expectation bias from Lemma 11.

The two direct held averages ν_H and H_O have means ν and s . Each changes by at most $2/(qD)$ per exchange, so the same argument, without extension, bounds both deviations by e_H .

There are $B_* = 2L(L-1) + 4$ statistics: one survival fraction, $L(L-1)$ weighted single-path quantities, $L(L-1)$ pair quantities, one mixed quantity, and two held averages. A union bound, including $\mathcal{G}^c$, gives equation 39. No conditional independence after imposing the degree event is used: concentration is applied to the globally defined extension, and equality on the degree event is used only afterward. $\square$

A.5.6 NORMALIZING THE MOMENTS AND COLLECTING THE CUBIC

The remaining step in the probability argument is deterministic. It divides by the same random survival fraction everywhere, then bounds products on the already constructed event.

Lemma 15 (Control of the cubic coefficients). *On the event $\mathcal{E}$ of Lemma 14, assume $e_0 < 1$. Then the normalized moments satisfy*

$$\begin{aligned} |K_{\ell\ell} - \nu| &\leq k_D, & |U_{\ell\ell} - \nu/(pL)| &\leq u_D, & |H_{\ell\ell} - \nu| &\leq e_H, \\ |K_{j\ell} - s| &\leq k_O, & |U_{j\ell} - s/L| &\leq u_O, & |H_{j\ell} - s| &\leq e_H, & |Z_{j\ell} - s/L| &\leq z_O \quad (j \neq \ell). \end{aligned} \quad (40)$$

Consequently the coefficients in equation 27 obey

$$|A_S - A_o| \leq E_A, \quad |B_S - B_o| \leq E_B, \quad |C_S - C_o| \leq E_C. \quad (41)$$

If also $A_- = A_o - E_A > 0$, then $A_S \geq A_-$ and

$$|B_S/A_S - u| \leq d_u, \quad |C_S/A_S - v| \leq d_v. \quad (42)$$

Proof. Divide by the random survival fraction. On $\mathcal{E}$, write $z = F_0/p^L$. Then $|z - 1| \leq e_0$. The scaled weighted single-path, diagonal-pair, off-diagonal-pair, and mixed statistics are within e_0, e_D, e_O, e_Z of their respective unmasked reference means. The identities equation 31–equation 32, together with $z \geq 1 - e_0$, give equation 40. For example, the pair identities divide by z^2 , and $|z^2 - 1| \leq 2e_0 + e_0^2$, giving equation 56.

Collect products on the same event. If $|x - x_0| \leq a$ and $|y - y_0| \leq b$, then $|xy - x_0y_0| \leq |x_0|b + |y_0|a + ab$. Apply this deterministic product inequality to every term of equation 27, using $Z_{\ell\ell} = 0$. There is one diagonal and $L - 1$ off-diagonal terms per role. This gives equation 41. Finally,

$$\left| \frac{B_S}{A_S} - u \right| \leq \frac{|B_S - B_o| + u|A_S - A_o|}{A_-},$$

and the same argument with C_S, v proves equation 42. This step multiplies simultaneous deterministic bounds on the same probability event. It does not factor an expectation over dependent mask statistics. $\square$

A.6 JOINT FACTOR DYNAMICS AND THE FINITE-TIME REMAINDER

We now connect the scalar cubic to actual training. The random-mask result above is used only through its candidate-marginal and coefficient bounds. All following trajectory estimates are for jointly trained w, b, C , not direct optimization of the product G .

Lemma 16 (Raw gradients and global existence). *Fix a mask with a nonempty reasoning population, and write $G_{r,e} = w_r b_{r,e}$ and $R_\ell = f_\ell - y_{a_\ell}$. The actual unit-rate gradients are*

$$Q_a := \partial_{G_a} F_\beta = \beta \sum_j C_{j1} \mathbb{E}_{\text{at}}[\mathbf{1}\{a_{1j} = a\} R_1] + \frac{1 - \beta}{L} \sum_{\ell,j} C_{j\ell} \mathbb{E}_S[\mathbf{1}\{a_{\ell j} = a\} R_\ell], \quad (43)$$

$$\dot{b}_{r,e} = -w_r Q_{r,e}, \quad \dot{w}_r = - \sum_e b_{r,e} Q_{r,e}, \quad (44)$$

$$\dot{C}_{j\ell} = -\beta \mathbf{1}\{\ell = 1\} \mathbb{E}_{\text{at}}[G_{a_{1j}} R_1] - \frac{1 - \beta}{L} \mathbb{E}_S[G_{a_{\ell j}} R_\ell]. \quad (45)$$

For every real $\beta \in [0, 1]$, this system has a unique solution for all $t \geq 0$ and smooth dependence on β and the initial parameters on finite time intervals.

Proof. Differentiate before taking the finite-population averages. Differentiating the common scalar predictor gives

$$\delta f_\ell = \sum_j C_{j\ell} \delta G_{a_{\ell j}} + \sum_j G_{a_{\ell j}} \delta C_{j\ell}, \quad \delta G_{r,e} = b_{r,e} \delta w_r + w_r \delta b_{r,e}.$$

Insert this identity into $\delta(R_\ell^2/2) = R_\ell \delta f_\ell$ and apply the two population weights in equation 4. This proves the displayed gradients. Atomic supervision contributes to every first-column gate parameter, and repeated candidate labels remain repeated terms.

Exclude finite-time blow-up. For real β , combine the two finite weighted populations into one Euclidean prediction vector f and target vector y . Its weights sum to one and $\|y\|^2 \leq 1$. In the following identity, the parameter norms are Euclidean for w and Frobenius for b, C . Trilinear homogeneity gives

$$\frac{d}{dt}(\|w\|^2 + \|b\|^2 + \|C\|^2) = -6\langle f, f - y \rangle = -6\|f - y/2\|^2 + \frac{3}{2}\|y\|^2 \leq \frac{3}{2}.$$

The polynomial vector field therefore cannot escape to infinity in finite time. It has a unique global real solution and smooth finite-time dependence on β and initialization. $\square$

Lemma 17 (The first nonzero held-learning coefficient). *For the raw flow with $w(0) = 1$, $b(0) = 0$, and $C(0) = E_{11}$, the first nonconstant held-evaluation coefficient is*

$$\mathcal{E}_H(\theta_{\beta,0}(t)) = \nu_H/2 - \frac{t^3}{2LD^2} P_S(\beta) + O(t^4). \quad (46)$$

The polynomial P_S is the complete signed cubic from equation 27. This identity holds for every fixed mask with a nonempty reasoning population.

Proof. Initial table motion. At $b(0) = 0$,

$$(Db_a)'(0) = [\beta + (1 - \beta)\omega_a]y_a =: V_a, \quad w'(0) = C'(0) = 0. \quad (47)$$

Only C_{11} is initially active, and the first atomic pair is uniform, which explains the first term.

Later routing starts at second order. For $\ell \geq 2$, equation 45 has no direct atomic term and yields

$$DC_{j\ell}''(0) = \frac{1 - \beta}{L} [\beta K_{j\ell} + (1 - \beta)U_{j\ell}].$$

Project the cubic prediction onto the held target. Later predictions first appear at order three. Their scalar target projections use $\mathbb{E}_{H,\ell}(V_{a_{\ell j}}y_{a_{\ell}}) = \beta H_{j\ell} + (1 - \beta)Z_{j\ell}$. Substitution proves equation 46. The squared-prediction term begins at order six. The next argument bounds the remainder and its mixture derivatives uniformly, including nonzero score initialization. $\square$

Lemma 18 (Blockwise analytic majorant). *Fix a mask whose candidate marginal satisfies equation 30, and use K, V, a, U, c, R_0 from Appendix A.9.2. For $\text{dist}(\beta, [0, 1]) \leq \rho$, the raw equations have a holomorphic time-series continuation for $|t| < R_0$. Coefficientwise absolute values are bounded by the positive system equation 60; in particular,*

$$\|b(t, \beta)\|_{\infty} \leq a(|t|), \quad \|w(t, \beta)\|_{\infty} \leq U(|t|), \quad \|C(t, \beta)\|_1 \leq c(|t|).$$

Here $\|C\|_1 = \sum_{j,\ell} |C_{j\ell}|$. For every later role $\ell \geq 2$,

$$|f_{\ell}(t, \beta)| \leq H_{\sigma}(|t|) = \frac{1 + \rho}{V} U(|t|) a(|t|) [c(|t|) - 1].$$

At $\sigma = 0$, the first nonzero coefficient of this last majorant is $h_{03}t^3$, with h_{03} as in equation 61.

Proof. Bound the three blocks separately. Fix a mask with candidate marginal at most J_+ . For complex β within distance ρ of $[0, 1]$, write $|\beta| \leq \beta_0 + \rho$ and $|1 - \beta| \leq 1 - \beta_0 + \rho$ for some real $\beta_0 \in [0, 1]$. Write $c_1 = \sum_j |C_{j1}|$. If $\|w\|_{\infty} \leq U$, $\|b\|_{\infty} \leq a$, and $\|C\|_1 \leq c$, then $|f_{\ell}| \leq Uac$ and $|R_{\ell}| \leq 1 + Uac$. The atomic candidate marginal is exactly $1/D$. Hence

$$|Q_a| \leq \frac{1 + Uac}{D} \left(|\beta|c_1 + \frac{J_+}{L} |1 - \beta|c \right) \leq \frac{K}{D} c(1 + Uac).$$

The overlapping two-source terms in the first column are added before the bound is taken. Equations equation 44–equation 45 now imply

$$\|\dot{b}\|_{\infty} \leq (K/D)Uc(1 + Uac), \quad \|\dot{w}\|_{\infty} \leq (NK/D)ac(1 + Uac), \quad \|\dot{C}\|_1 \leq VUa(1 + Uac).$$

This proves the positive majorant system equation 60.

Obtain an explicit analytic radius. On writing $a = a_0z$ and $s = \gamma t$ only for this bounding system, its equations become

$$z' = Uc(1 + a_0Uzc), \quad c' = Uz(1 + a_0Uzc), \quad U' = Na_0^2zc(1 + a_0Uzc).$$

The invariants in Appendix A.9.2 follow by differentiation. Before $z = z_*$, they give $U \leq U_*$, $c \leq c_*$ and $z' \leq U_*c_*(1 + a_0U_*z_*c_*)$. Thus the positive solution exists up to R_0 in physical time. This does not rescale or freeze any model parameter.

Justify complex continuation, not only a real bound. Expand the raw polynomial equations formally in powers of t . The coefficient of order $n + 1$ is determined by orders at most n , divided by $n + 1$. Expand the finite-population expectations before taking absolute values. The block bounds above then give, inductively, coefficientwise domination by the positive majorant, uniformly over the complex mixture neighborhood. Complexification uses the same algebraic products as the real equations, without introducing complex conjugates.

It remains to show that these formal series converge. Every coefficient of the majorant is nonnegative. A finite partial series is a subsolution of the cooperative polynomial system: differentiation retains its lower-order coefficients, while the omitted right-hand-side coefficients are nonnegative. Comparison with the bounded positive solution therefore bounds every partial sum below R_0 .

The increasing partial sums converge, and coefficientwise domination gives absolute uniform convergence of the raw series on every smaller complex-time disk. The coefficients are polynomial in β , since the vector field is polynomial and its mixture weights are affine. Thus the sum is holomorphic in both variables; local analytic uniqueness identifies it with the raw flow. The strict inequality $\tau < R_0$ permits slightly enlarged closed contours for the Cauchy estimates below. A real-trajectory bound alone would not justify this step.

Use the smaller bound for later query columns. For each later role, the sum of its routing coefficients has physical derivative majorized by $(1+\rho)Ua(1+Uac)$. It starts at zero. Comparison with $c' = VUa(1+Uac)$ bounds that column by $(1+\rho)(c-1)/V$, coefficientwise. Its prediction is therefore majorized by $H_\sigma(t)$ in equation 61. At $\sigma = 0$ the first nonzero coefficient is exactly $h_{03}t^3$ in this majorant. $\square$

Lemma 19 (Finite-time remainder for joint training). *On $\mathcal{E}$, assume $A_- > 0$ and $0 < \tau < R_0$, and use the general initialization equation 11. The exact normalized gain has the representation*

$$-\frac{2LD^2}{\tau^3 A_S}[\mathcal{E}_H(\theta_\beta(\tau)) - \nu_H/2] = \frac{P_S(\beta)}{A_S} + r_{\text{time}}(\beta), \quad \|r_{\text{time}}^{(k)}\|_\infty \leq T_k \quad (k = 1, 2). \quad (48)$$

The uniform norms are over $\beta \in [0, 1]$ and T_1, T_2 are the explicit bounds in equation 61. The cubic P_S is computed with $B(0) = 0$, while r_{time} includes both higher time orders and the effect of nonzero $B(0)$.

Proof. Separate initialization terms coefficientwise. Let $f_{\ell,\sigma}$ and $f_{\ell,0}$ be the predictions with score initialization σZ and zero. Expand their formal coefficients also as polynomials in the initial score entries. Subtracting the zero-score series removes precisely the terms of degree zero in those entries. Applying the positive-coefficient argument to the remaining terms gives

$$|f_{\ell,\sigma}(\tau, \beta) - f_{\ell,0}(\tau, \beta)| \leq H_\sigma(\tau) - H_0(\tau)$$

uniformly in the complex mixture neighborhood. This assertion follows coefficientwise: one never subtracts two absolute-value inequalities.

Remove the exact cubic coefficient. Likewise, subtracting the actual cubic coefficient from $f_{\ell,0}$ leaves only orders at least four, bounded by $H_0(\tau) - h_{03}\tau^3$. Therefore

$$|f_{\ell,\sigma}(\tau, \beta) - [t^3]f_{\ell,0}(t, \beta)\tau^3| \leq H_\sigma(\tau) - h_{03}\tau^3. \quad (49)$$

The right side is nonnegative. This step retains the effects of nonzero score initialization; it does not set σ to zero in the theorem.

Pass from predictions to the evaluation loss. Using the scalar identity $(f - y)^2/2 = y^2/2 - yf + f^2/2$, equation 46, and equation 49,

$$\left| \mathcal{E}_H(\theta_\beta(\tau)) - \nu_H/2 + \frac{\tau^3}{2LD^2}P_S(\beta) \right| \leq m_H\{H_\sigma(\tau) - h_{03}\tau^3\} + \frac{1}{2}H_\sigma(\tau)^2 \leq \mathcal{T}. \quad (50)$$

Here $m_H = \mathbb{E}_{H,\ell}|y_{a_\ell}| \leq \min\{1, \sqrt{\nu_H}\} \leq m_+$ on $\mathcal{E}$.

Control the first two mixture derivatives. The expression is holomorphic in β by Lemma 18. Cauchy's derivative formula on a circle of radius ρ around each real β gives derivative bounds $k!\rho^{-k}\mathcal{T}$. Dividing by $\tau^3 A_S/(2LD^2)$ and using $A_S \geq A_-$ proves the uniform derivative bounds T_1, T_2 in equation 61. This proves equation 48. No high-degree polynomial root or learned Hessian is needed for this estimate. $\square$

A.7 FROM THE TWO ERROR BOUNDS TO THE TRAINED OPTIMUM

The preceding results control the normalized coefficients and the finite-time remainder. The next deterministic lemma explains exactly how those controls imply a bound on the optimizer; it is the only optimization step needed in the final proof.

Lemma 20 (Deterministic optimizer stability). *Let R_o have the center and margins of Lemma 3, and suppose an exact normalized gain on $[0, 1]$ has the form*

$$J(\beta) = R_o(\beta) + \Delta u \beta(1 - \beta)^2 + \Delta v (1 - \beta)^3 + r(\beta),$$

where $|\Delta u| \leq d_u$, $|\Delta v| \leq d_v$, $\|r'\|_\infty \leq T_1$, and $\|r''\|_\infty \leq T_2$. Set $d_1 = d_u + 3d_v + T_1$ and $d_2 = 2d_u + 4d_v + T_2$. If $d_1 < m_o$ and $d_2 < \mu_o$, then J has a unique global maximizer $\beta^* \in (1/3, 2/3)$, and

$$|\beta^* - \Theta| \leq \frac{d_u(1 - \Theta)(3\Theta - 1) + 3d_v(1 - \Theta)^2 + T_1}{\mu_o - d_2}.$$

The curvature assumption concerns the explicit reference cubic through its margins, not a Hessian at an unknown trained solution.

Proof. Locate every global maximizer. On $[0, 1]$ the first derivatives of the two displayed basis polynomials have respective absolute bounds 1 and 3. On $[1/3, 2/3]$, their second derivatives have absolute bounds 2 and 4. Thus the gain's first derivative differs from R'_o by at most d_1 , and its second derivative differs on the central interval by at most d_2 .

Lemma 3 and the two strict error inequalities now make the exact gain strictly increasing on $[0, 1/3]$, strictly decreasing on $[2/3, 1]$, and strictly concave on the central interval. A global maximizer exists by continuity and compactness; the exterior signs place every maximizer in $(1/3, 2/3)$ and the central strict concavity makes it unique.

Bound the displacement from the known center. At the known analytic center Θ , $R'_o(\Theta) = 0$. The gain derivative there has magnitude at most

$$d_u(1 - \Theta)(3\Theta - 1) + 3d_v(1 - \Theta)^2 + T_1.$$

On the whole central interval its negative second derivative is at least $\mu_o - d_2$. The mean-value theorem between Θ and the unique maximizer proves the stated displacement bound. $\square$

Proof of Theorem 2. Choose the common probability event. Lemma 14 gives $\Pr(\mathcal{E}) \geq 1 - \delta$. On this event the reasoning population is nonempty by Lemma 6. Lemma 15 gives $A_S \geq A_- > 0$ and

$$\Delta u = B_S/A_S - u, \quad \Delta v = C_S/A_S - v, \quad |\Delta u| \leq d_u, \quad |\Delta v| \leq d_v.$$

Apply the deterministic stability lemma. Lemma 19 represents the normalized exact gain as

$$R_o(\beta) + \Delta u \beta(1 - \beta)^2 + \Delta v (1 - \beta)^3 + r_{\text{time}}(\beta).$$

All hypotheses of Lemma 20 are precisely the final two admissibility conditions in equation 64. Its conclusion is equation 15, including uniqueness and the interval $(1/3, 2/3)$. The gain is the negative of a positively rescaled evaluation error after subtracting a mixture-independent baseline, so its maximizer is exactly $\beta_{\tau, S}^*$. $\square$

A.8 DEDUCTION OF THE MAIN-TEXT ASYMPTOTIC THEOREM

The finite-dimensional result keeps the constants explicit. The following lemma identifies their order in the regime used by the main text.

Lemma 21 (Asymptotic radius). *Fix $L, p, q, \delta, \rho, \tau$, set $B(0) = 0$, and let $M/\log(ND) \rightarrow \infty$ with $2 \leq M \leq N$ and $D = NM$. For sufficiently large dimensions all conditions in equation 64 hold, and the explicit radius satisfies*

$$\mathfrak{R} = O_{L,p,q,\rho,\tau} \left(\frac{1}{M} + \sqrt{\frac{\log(1/\delta)}{NM}} + \frac{1}{NM} \right).$$

Proof. Random-mask errors. Fix $L, p, q, \delta, \rho, \tau$, the initialization $B(0) = 0$, and $M/\log(ND) \rightarrow \infty$. Then $h \rightarrow 0$, so α_h, Λ_h, J_+ are bounded. The collision terms in equation 54 are $O(1/M)$, the finite-population deficits are $O(1/D)$, and every concentration term is $O(\sqrt{\log(1/\delta)/D})$. Thus E_A, E_B, E_C and d_u, d_v have the mask-error order in equation 8. Since $\nu \geq 1/4$, $A_\circ \geq 1/16$; for fixed L, p , the positive margins of Lemmas 3 and 4 are uniformly bounded below over $\kappa \in [0, 1]$.

Finite-time error at a fixed physical horizon. For the time contribution, the positive equations give, uniformly on a fixed time interval,

$$\begin{aligned} a(t) &= Kt/D + O((t^2 + t^3)/D^2), \\ c(t) - 1 &= VKt^2/(2D) + O((t^3 + t^4)/D^2), \\ U(t) - 1 &= NK^2t^2/(2D^2) + O(D^{-2}), \end{aligned}$$

where $M \geq 2$ makes $N/D \leq 1/2$ and the constants may depend on the fixed time interval. These follow by one substitution into the integral equations, using a first-exit bound $a = O(D^{-1})$, $c - 1 = O(D^{-1})$, $U - 1 = O(D^{-1})$ there. Hence $H_0(t) - h_{03}t^3 = O((t^4 + t^5)/D^3)$ and $H_0(t)^2 = O(t^6/D^4)$. Equation equation 61 gives $T_k = O((\tau + \tau^2)/D)$ for fixed τ , yielding equation 8. The explicit radius equation 62 grows proportionally to $\sqrt{D}$ in this regime, so every fixed positive τ is eventually admissible.

Admissibility and separated comparisons. Strict center gaps between fixed separated settings have a positive lower bound over the compact range $\kappa \in [0, 1]$, and the radii vanish. Thus all admissibility conditions hold for sufficiently large dimensions. $\square$

Proof of Theorem 1. Use $\sigma = 0$ in Theorem 2, which is exactly the main-text initialization equation 6. Since $ND = N^2M$, the dimensional condition in Lemma 21 is the condition in the main theorem. That lemma verifies admissibility for all sufficiently large dimensions and supplies the stated rate. Theorem 2 then gives nonemptiness, uniqueness, and the probability bound. Lemma 4 gives both center derivatives. $\square$

A.9 EXPLICIT CONSTANTS AND ADMISSIBILITY

All bounds below are dimension-only sufficient bounds. None is estimated from a realized mask. Fix the same auxiliary choice $\rho > 0$ throughout. Lemmas 14, 15, and 19 justify their respective roles in the proof.

A.9.1 DEGREE ENVELOPE AND RAW-MOMENT ERRORS

Let $D = NM$, $q = 1 - p$, and put

$$\begin{aligned} \delta_g &= \frac{\delta q p^{2L}}{16D}, & h &= \sqrt{\frac{3 \log(4N/\delta_g)}{pM}}, & 0 < h < 1, \\ \alpha_h &= (1 + h)^{L-1}, & \Lambda_h &= \left(\frac{1 + h}{1 - h} \right)^{L-1}, & J_+ &= \frac{\Lambda_h}{p(1 - h)}. \end{aligned} \tag{51}$$

We do not require $h \leq 1/(4L)$. Define

$$\begin{aligned} c_0 &= \frac{2L\alpha_h}{pD}, & c_D &= \frac{2(2L-1)\alpha_h^2}{pD}, \\ c_O &= \frac{4L\alpha_h^2}{p^2D}, & c_Z &= \frac{2\alpha_h}{qD} \left(\frac{L}{p} + 1 + h \right). \end{aligned} \tag{52}$$

These constants control raw path sums on the degree event, not ratios of conditional expectations. With

$$B_* = 2L(L-1) + 4, \quad \ell_\delta = \log(4B_*/\delta), \quad v_\delta = \sqrt{\frac{\min\{p, q\}D\ell_\delta}{2}}, \quad e_{\text{ext}} = \frac{\delta}{8D}, \quad (53)$$

define the finite-population deficit and collision factors

$$\begin{aligned} f_k &= \frac{k(k-1)}{2pD}, \quad t_p = \frac{q}{p}, \\ b_{\text{self}} &= \frac{(1+t_p L/M)^{L-1} - 1}{M}, \quad F_*^{\text{col}} = (1+2t_p L/M)^{L-1}, \\ b_D &= (1+b_{\text{self}})F_*^{\text{col}} - 1, \quad b_O = (1+b_{\text{self}}/p + t_p L/M)F_*^{\text{col}} - 1, \\ b_Z &= \frac{1}{M} + \frac{L-1}{M^2} + f_L + \frac{b_{\text{self}} + L/D}{1-L/D}. \end{aligned} \quad (54)$$

The requirement $pD \geq 2L$ makes the falling-factorial bounds used here valid. Set

$$\begin{aligned} e_0 &= b_{\text{self}} + f_L + e_{\text{ext}} + c_0 v_\delta, \\ e_D &= b_D + f_{2L-1} + e_{\text{ext}} + c_D v_\delta, \\ e_O &= b_O + f_{2L} + e_{\text{ext}} + c_O v_\delta, \\ e_Z &= b_Z + e_{\text{ext}} + c_Z v_\delta, \quad e_H = \frac{2v_\delta}{qD}. \end{aligned} \quad (55)$$

Here e_0 denotes a normalization-error bound, not the chain's initial entity. Require $e_0 < 1$. The errors after dividing by the random valid-chain normalization are

$$\begin{aligned} k_D &= \frac{(1+\nu)e_0}{1-e_0}, \quad k_O = \frac{(1+s)e_0}{1-e_0}, \\ u_D &= \frac{e_D + \nu(2e_0 + e_0^2)}{pL(1-e_0)^2}, \quad u_O = \frac{e_O + s(2e_0 + e_0^2)}{L(1-e_0)^2}, \\ z_O &= \frac{e_Z + se_0}{L(1-e_0)}. \end{aligned} \quad (56)$$

For nonnegative x, y, a, b , write $\Pi(x, y; a, b) = xb + ya + ab$. The three cubic-coefficient errors are

$$\begin{aligned} E_A &= \Pi(\nu, \nu; k_D, e_H) + (L-1)\Pi(s, s; k_O, e_H), \\ E_B &= \Pi(\nu/(pL), \nu; u_D, e_H) \\ &\quad + (L-1)\{\Pi(s/L, s; u_O, e_H) + \Pi(s, s/L; k_O, z_O)\}, \\ E_C &= (L-1)\Pi(s/L, s/L; u_O, z_O). \end{aligned} \quad (57)$$

To define the normalized errors and the reference margins, put

$$A_- := A_\circ - E_A, \quad d_u := \frac{E_B + uE_A}{A_-}, \quad d_v := \frac{E_C + vE_A}{A_-}, \quad (58)$$

$$m_\circ := \min\{u - 3v, (1-4v)/3, (u+v)/3, 1\}, \quad \mu_\circ := 2\min\{u - 2v, 1 - v\}.$$

The admissibility conditions below require $A_- > 0$. The margins are strictly positive by Lemma 3.

A.9.2 A SMALL THREE-BLOCK BOUND FOR PHYSICAL TRAINING TIME

Set $m_+ = \min\{1, \sqrt{\nu + e_H}\}$ and

$$K = \max\{1, J_+/L\} + \rho(1 + J_+/L), \quad V = L(1 + 2\rho). \quad (59)$$

Let $a(t), U(t), c(t)$ be the positive solution of the three scalar equations

$$\begin{aligned} a' &= \frac{K}{D}Uc(1+Uac), \quad a(0) = \sigma, \\ U' &= \frac{NK}{D}ac(1+Uac), \quad U(0) = 1, \\ c' &= VUa(1+Uac), \quad c(0) = 1. \end{aligned} \quad (60)$$

This is a known bounding system, not a learned trajectory. It has only three scalar coordinates, independent of the model dimension. Define

$$\begin{aligned} H_\sigma(t) &= \frac{1+\rho}{V} U(t) a(t) [c(t) - 1], & h_{03} &= \frac{(1+\rho)K^2}{2D^2}, \\ \mathcal{T} &= m_+ \{H_\sigma(\tau) - h_{03}\tau^3\} + \frac{1}{2} H_\sigma(\tau)^2, \\ T_k &= \frac{2LD^2}{A_- \tau^3} \frac{k!}{\rho^k} \mathcal{T}, & k &= 1, 2. \end{aligned} \quad (61)$$

The expression in braces is nonnegative. To specify a sufficient existence and complex-analytic radius explicitly, set

$$\begin{aligned} a_0 &= \sqrt{K/(DV)}, & \gamma &= \sqrt{KV/D}, & z_0 &= \sigma/a_0, & z_* &> z_0, \\ U_* &= \sqrt{1 + Na_0^2(z_*^2 - z_0^2)}, & c_* &= \sqrt{1 + z_*^2 - z_0^2}, \\ R_0 &= \frac{z_* - z_0}{\gamma U_* c_* (1 + a_0 U_* z_* c_*)}. \end{aligned} \quad (62)$$

Require $0 < \tau < R_0$ for some choice of z_* . The invariants

$$U(t)^2 - Na(t)^2 = 1 - N\sigma^2, \quad c(t)^2 - \frac{VD}{K} a(t)^2 = 1 - \frac{VD}{K} \sigma^2$$

also reduce evaluation of equation 60 to a single increasing scalar equation, or a one-dimensional integral. No tensor of training moments is needed to evaluate this bound.

A.9.3 ADMISSIBILITY AND THE EXPLICIT THEOREM RADIUS

For the chosen auxiliary radius ρ and a choice $z_* > z_0$ in equation 62, define

$$d_1 := d_u + 3d_v + T_1, \quad d_2 := 2d_u + 4d_v + T_2. \quad (63)$$

The finite-sample, time, and initialization conditions used in Theorem 2 are precisely

$$\begin{aligned} pD &\geq 2L, & 0 &< h < 1, & e_0 &< 1, & A_- &> 0, \\ 0 &< \tau < R_0, & d_1 &< m_o, & d_2 &< \mu_o. \end{aligned} \quad (64)$$

All displayed quantities must be well-defined. These are deterministic conditions on $N, M, L, p, \delta, \tau, \sigma$ and the auxiliary choices; they do not involve a realized mask, a trained optimizer, or its Hessian. The radius in equation 15 is

$$\mathfrak{R} := \frac{d_u(1 - \Theta)(3\Theta - 1) + 3d_v(1 - \Theta)^2 + T_1}{\mu_o - d_2}, \quad \Theta = \Theta_{N,M}(L, p). \quad (65)$$

The admissibility conditions ensure that the normalizations are defined and the denominator of equation 65 is positive. For the rate in equation 8, all auxiliary parameters are held fixed; the implicit constants may depend on L, p, q, ρ, τ .